

COMP ENG 3DQ5 - Project Report

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Group 84

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1. Introduction

The primary goal of this project is to implement an image decompressor in hardware using processes such as lossless decoding, dequantization, signal transformation, interpolation, and color space conversion. The approach is based on reverse engineering, starting with interpolation and color space conversion, progressing to the Inverse Discrete Cosine Transform, and concluding with dequantization and lossless decoding. These objectives provide insight into the course's broader concepts. The Verilog design is tested with a testbench, simulated using Quartus to evaluate timing and circuit behavior, and deployed onto an FPGA for practical application.

2. Implementation Details

2.1. Upsampling and Colour Space Conversion (Milestone 1)

	0	1	2	3	4	5	6
	E	E	U	V	O	O	X
	E	E	U	V	O	O	X
	E	X	U	V	O	X	X

The figure above represents the overall design for Milestone 1. The implementation works as follows: colour space conversion calculations are done for the even pixel, interpolation for U and V are then used to calculate the colour space conversion for the odd pixel. The calculations are done using the previous U and V shifts. In the common case, U and V values are fetched every 12 clock cycles. RGB values are calculated every 7 clock cycles, during which 3 writes are completed and the RGB values are written to the SRAM.

Module	Register Name	Bits	Description
M1	Y_SEGMENT_counter U_SEGMENT_counter V_SEGMENT_counter RGB_SEGMENT_counter	18	Used to properly address where we fetch and write data from/to SRAM.
M1	Counter	18	Counter is used to increment the counters used for addressing.
M1	Counter2	18	Counter2 is used as a pixel counter and is incremented every time Y is fetched.
M1	Stop_counter	18	Used to stop the module and go back to the project.sv.
M1	Y_buffer	16	Temporarily stores Y data fetched from SRAM.
M1	U_buffer_odd V_buffer_odd	32	Stores the odd pixel that isn't being currently shifted in.

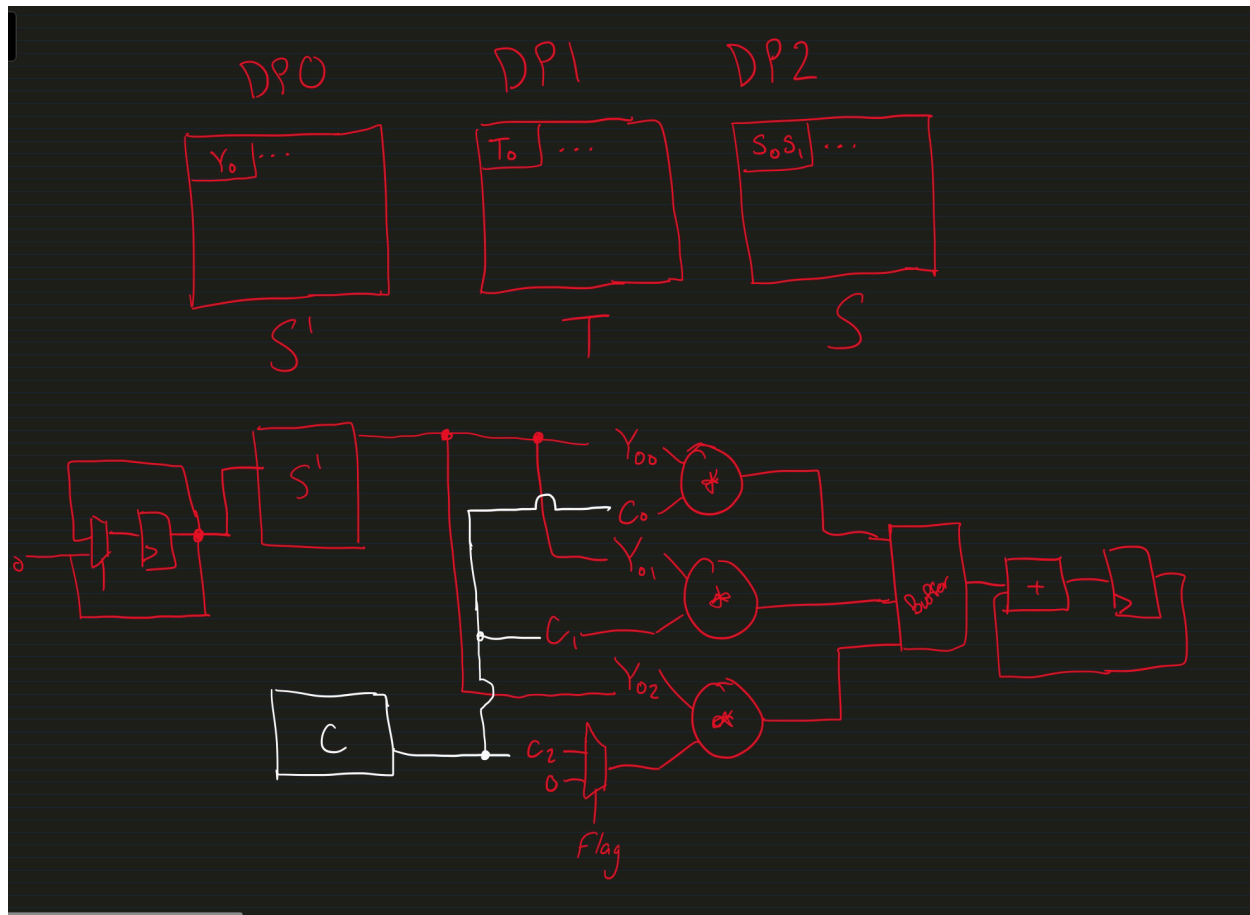
M1	shift1 - shift6 vshift1 - vshift6	32	Used to hold fetched U and V values used to interpolate the odd pixel.
M1	adder1 adder2 adder3	32	Holds the result of the additions.
M1	m1 m2 m3	32	Temporary buffer to store the results of the initial 3 multiplication (m1_result, m2_result, and m3_result).
M1	m1_coeff m2_coeff m3_coeff	32	Holds the predefined constants for the multipliers.
M1	m1_term m2_term m3_term	32	Holds calculations for multiplier use.
M1	m1_result_long m2_result_long m3_result_long	64	These are the pre-clipped versions of the multiplication.
M1	m1_result m2_result m3_result	32	These registers hold the multiplier result and clip the long version back to 32 bits from 64 bits.
M1	inter_U_buffer inter_V_buffer	32	Stores U and V interpolation results for multiplier use.
M1	inter_U inter_V	32	Hold interpolated U and V values after calculations.
M1	R_even G_even B_even R_odd G_odd B_odd	32	Holds calculated R, G, and B values for even and odd pixels.
M1	red_even red_odd green_even green_odd blue_even blue_odd	8	Clipped RGB values for even and odd pixels.

$$\begin{aligned}
 & (14 \text{ clock cycles in common case} * 160 \text{ columns}) + (1 \text{ leadout cycle} + 15 \text{ leadin cycles}) * 240 \\
 & \quad \text{rows} \\
 & = 541,440 \text{ total clock cycles to complete Milestone 1}
 \end{aligned}$$

$$\begin{aligned}
 & (32 \text{ multipliers calculations used in common case} / 42 \text{ common cases}) \\
 & = \sim 76\% \text{ utilization}
 \end{aligned}$$

In the common state we use 14 clock cycles, with a constraint of 3 multipliers with at least 75% utilization. During which in the common state there are only 10 times out of the 42 times where the multipliers are not in use. Making our utilization around 76%.

2.2. Inverse Discrete Cosine Transform (Milestone 2)



Module	Register Name	Bits	Description
M2	write_s0 write_s1 write_s2	8	Used to clip the S value
M2	Counter	18	Counter is used to increment the counters used for addressing.
M2	Counter2	18	Counter2 is used as a pixel counter and is incremented every time Y is fetched.
M2	dp_ram_address_counter dp_ram_address_counter_2	18	Used to address the dp rams.
M2	row_counter col_counter	18	Used to indicate row and column within a matrix.
M2	block_counter	18	Tracks which block it's at

M2	c0 c1 c2	32	Holds the C matrix values
M2	m1_term m2_term m3_term	32	Holds splicing versions of Y_buff_X (16 bits)
M2	Y_buff_1 Y_buff_2	64	Holds value pre spliced value of S`
M2	T_buff_sum	32	Store the sum of T after matrix calculation
M2	C0 C1 C2	32 (8x8)	Predefined C matrix
M2	CT	32 (8x8)	Transpose C matrix

2.3. Lossless Decoding and Dequantization (Milestone 3)

N/A as we didnt get to milestone 3.

2.4. Resource Usage and Critical Path

When we checked the Timing Analyzer, we saw that all slack values were positive, meaning our design has no timing problems. The longest delay is 3.609ns, happening from milestone 1 inter_V_buffer. The estimated total logic elements within the project is 2012.

3. Weekly Activity and Progress

Week/Date	Project Progress	Individual contribution
Week 1 (Oct 27th)	Began on reading the project document, and trying to understand what the project (M1) was about. Started on creating the state table for milestone 1.	Chris read the project document and asked clarifying questions to TA's to understand what was going on and what needed to be done. Chris started working on the state table and brainstormed on the design to be implemented. Sohel read the project document and attended the first lecture on the project to understand what was going on.
Week 2 (Nov 3rd)	Finished the state table, and got approval by TA. Began Coding and implementing our state table.	Chris started coding the leadin state common and leadout from the state table and realized that our design needed to be changed. Sohel implemented the leadout state and assisted/edited the

		common state to which we had to accommodate the design change for.
Week 3 (Nov 10th)	Changed up the state table after realizing changes need to be made. Finished up most of the code which allowed us to start debugging in modelsim.	Chris made the necessary changes to the leadin state and when that started working as intended he applied those changes to the common state and leadout state. Chris started debugging, going through and fixing one error at a time. Sohel coded the leadout state to which it was redesigned due to it not working as intended. Sohel Assisted with the debugging by pair coding which was hand calculating values in the modelsim, and analyzing logic/syntax errors.
Week 4 (Nov 17th)	Finished milestone1 where it successfully ran on the board and displayed the image. Began on milestone 2.	After making sure our M1 worked on the board we started looking into M2 by understanding what needed to be done. Chris started working on Fetch S', calc S and write S. Sohel started working on Compute T.
Week 5 (Nov 24th)	Worked on milestone 2 and finished the report.	Both worked on report. Sohel focused on the report while Chris focused on milestone 2. Chris did a final look through of the report.

4. Conclusion

This class has provided more than just an understanding of digital system design, it has taught us how to think creatively and translate our ideas from conceptual designs into functional Verilog implementations. The structure of this course stood out from any other we have taken, as it required a significant investment of time and effort, challenging us in ways few courses have. The project has taught us essential skills like thoroughly analyzing project documents, developing designs, coding, and debugging effectively. Overall, we gained valuable hands-on experience that will benefit us in the future.

	Git message	Completion Date	Tasks finished
Milestone 1	“finished m1”	November 20, 2024	Everything
Milestone 2	Incomplete	Incomplete	Fetch S' (only implemented

			not debugged) Compute T (only implemented not debugged) Compute S (only implemented not debugged) Write S (only implemented not debugged)
Milestone 3	N/A	N/A	N/A

5. References

- [1] Nicolici, N., Thong, J., Kinsman, A. 2024. “COMP ENG 3DQ5 Project Description 2024: Hardware Implementation of an Image Decompressor” In: 3DQ5. Pages 1-31.